Mechanisms · Teacher and Cover Plan

Learning goal: I can identify levers, pulleys, and gears and explain how they let a small effort move a heavy load.

Scheme Outcome

TRACK Science scheme of work, Unit 1 (Forces, Mechanisms and Electricity), Topic: Recognise that some mechanisms, including levers, pulleys and gears, allow a smaller force to have a greater effect.

Identify levers, pulleys, and gears in everyday tools and explain how they allow a smaller effort force to produce a larger effect on a load.

Success Criteria

I can label the pivot, effort, and load on a simple lever diagram.

I can match everyday tools to levers, pulleys, or gears.

I can explain that a mechanism trades a longer push distance for a smaller effort force.

Model Sequence (I do)

First, look at the lever: it has a load, a pivot, and an effort push.

When the effort push is far from the pivot and the load is close, a small push lifts a big load.

Pulleys use a wheel and rope, and gears use interlocking teeth to change force, speed, or

direction.

Nothing is gained for free: if a lever halves the effort force, your hand must push twice as far.

A single fixed pulley only changes the direction of the pull (pulling down lifts the load up).

Adding a second movable pulley shares the load across more rope sections, reducing the effort force needed.

We Do & Hinge Check

A small gear with 10 teeth turns a larger gear with 20 teeth. What happens to the larger gear?

Answer: A. Meshed gears turn in opposite directions; driving a larger gear reduces speed and increases turning force.

Safe Practical & Card Setup

Explore the ruler lever, card gear pair, and pulley diagram cards at your desk.

Kit: 30 cm stiff ruler, triangular pencil/eraser pivot, small mass load, card gear wheels, and mat.

Keep the ruler lever low on the desk. Do not flick rulers or drop masses near fingers.

You Do Routes (Supported, Independent, Aspire)

Supported: Label Load, Pivot, and Effort on the lever diagram and match crowbar, flagpole, and bike cog to lever, pulley, and gear. (Help: Bar + pivot = lever. Wheel + rope = pulley. Toothed wheels = gears.)

Independent: Draw and label a lever (Load, Pivot,

Effort), classify four everyday tools as lever, pulley, or gear, and explain how a lever multiplies force. (Help: A lever uses a __ to turn a small __ force pushed over a long distance into a larger force on the __.)

Aspire: Compare how a 10-tooth gear driving a 30-tooth gear and a long-armed lever both trade distance/speed for greater force. (Help: The 10-tooth gear must turn __ times to turn the 30-tooth gear once, trading speed for __ turning force.)